

Linear Algebra Recitation

Week 14

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Singular Value Decomposition $A = WDW^T$ (if A is sym)

True or false: for any matrix A , the matrices $A^T A$, AA^T are symmetric.
 $\in \mathbb{R}^{m \times n}$ $\in \mathbb{R}^{n \times n}$ $\in \mathbb{R}^{m \times m}$

$$(A^T A)^T = A^T (A^T)^T = A^T A$$

$$(AA^T)^T = (A^T)^T A^T = AA^T$$

True or false: for any matrix A , the matrices $A^T A$, AA^T are diagonalizable.

spectral theory of sym. matrices.

Singular Value Decomposition

True or false: for any matrix A , the eigenvalues of $A^T A$, AA^T are non-negative. $A^T A x = \lambda x$, $\|x\| = 1$

$$\underline{\lambda} = \lambda \|x\|^2 = \lambda (x \cdot x) = (x \cdot \lambda x) = (x \cdot A^T A x) = (Ax \cdot Ax) = \|Ax\|^2$$

True or false: for any matrix A , the non-negative eigenvalues of $A^T A$, AA^T are the same. (cf) last week's slides

$$A^T A x = \lambda x$$

& quiz

$$AA^T Ax = \lambda Ax \quad AA^T y = \lambda y$$

What do you call the square roots of the eigenvalues of $A^T A$ or AA^T ?

Singular Values of A .

Singular Value Decomposition Eval decomp.

$$A = W D W^{-1}$$

The SVD of a matrix $A \in \mathbb{R}^{m \times n}$ is

$$A = U S V^T,$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are **orthogonal** matrices, and $S \in \mathbb{R}^{m \times n}$ contains the singular values on its diagonal. The columns of U are called the “left singular vectors” of A , and the columns of V are called the “right singular vectors” of A .

Question: The eigenvectors of $A^T A$ will become the (left/right) singular vectors of A . The eigenvectors of $A A^T$ will become the (left/right) singular vectors of A .

This is the reason we can find **orthogonal** matrices U, V .

Singular Value Decomposition

How to compute ^athe SVD $A = USV^T$.

- 1 Find e.val decomposition of $A^T A$ or AA^T . (Choose the one that is smaller or easier to compute) $\implies WDW^T$.
- 2 Set S to be the “square root” matrix of D . $\triangle S$ is not square.
- 3 Set W as either left or right singular vector (i.e., $W = U$ or V).
- 4 if $W = V$, find U via the identity $Av_i = \sigma_i u_i$. $AV = US$
- 5 if $W = U$, find V via $u_i^T A = \sigma_i v_i^T$. $U^T A = SV^T$

Note 1: once you get U by finding the eigenvectors of AA^T , you **don't** want to compute V by finding eigenvectors of $A^T A$. Instead, use the identity above.

Note 2: if there are singular values that are zero, this means the corresponding singular vectors can be any vector that make U and V orthogonal. Choosing the right vectors in this case require some practice.

Exercises

Easy.

Find ~~the~~ ^{two} SVD^s of $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

$$A^T A = A A^T = I \Rightarrow (A^T A - I) = 0$$

$$S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\downarrow$
Null space: $\mathbb{R}^2$

$$A = U S V^T = U I V^T$$

① $U = I, V = I \Rightarrow A = I I I = I$: Valid

② $U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow A = U I V^T = I$: Valid

③ $U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}, V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$: Valid

④ $U = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, V = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} : A = U I V^T = U^{-1}$
 $\nrightarrow$ not an SVD (U, V not orth)
but is an e.val decomp.

Exercises obvious: $A = I \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} I$ is a valid SVD.

Easy.

Find the SVD of $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

$$S = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

e. vec of $AA^T = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$$\sigma_1^2 = 1 \Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\sigma_2^2 = 0 \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$u_i^T A = \sigma_i v_i^T$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = v_1^T$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$u_2^T A = \sigma_2 v_2^T$$

What to do?

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

v_2 : chosen to make V orthogonal.

$$A = U S V^T$$

When A is diagonalizable, the SVD gives the e.val Decomposition.

$\hookrightarrow$ Singular values $\sigma_i = \text{e.vals } \lambda_i$

Exercises

Requires some calculation.

$$A^T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad AA^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \begin{matrix} \sigma_1^2 = 3 \\ \sigma_2^2 = 1 \end{matrix}$$

Find the SVD of $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. $(AA^T - 3I) = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \Rightarrow X_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

$$S = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$(AA^T - I) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow X_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$\frac{u_i^T A}{\sigma_i} = v_i^T$$

$$V = \begin{bmatrix} \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -1/\sqrt{2} & -1/\sqrt{3} \\ \frac{1}{\sqrt{6}} & 1/\sqrt{2} & -1/\sqrt{3} \end{bmatrix}$$

$\begin{matrix} \uparrow & \uparrow & \uparrow \\ v_1 & v_2 & v_3 \end{matrix}$

$$\left(\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right) \frac{1}{\sqrt{3}} A = \left(\frac{2}{\sqrt{6}} \quad \frac{1}{\sqrt{6}} \quad \frac{1}{6} \right) = v_1^T$$

$$\left(\frac{1}{\sqrt{2}} \quad -\frac{1}{\sqrt{2}} \right) A = \left(0 \quad -\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right) = v_2^T$$

$A = U S V^T$
 $2 \times 2 \quad 2 \times 3 \quad 3 \times 3$

$\rightsquigarrow v_3$? Choose any vector that makes V orthogonal: $v_3^T = \left[\frac{1}{\sqrt{3}} \quad \frac{1}{\sqrt{3}} \quad \frac{1}{\sqrt{3}} \right]$

Exercises

Requires some calculation.

$$A^T = \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$$

Find the SVD of $A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$. $AA^T = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}$ ← choose this one!

$$\sigma_1^2 = 8 \Rightarrow AA^T - 8I = \begin{bmatrix} 0 & 0 \\ 0 & -6 \end{bmatrix} \Rightarrow x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\sigma_2^2 = 2 \Rightarrow AA^T - 2I = \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, S = \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}$$

Use $u_i^T A = \sigma_i v_i^T$

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \frac{1}{2\sqrt{2}} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = v_1^T$$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = v_2^T$$

$$\Rightarrow V = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$\uparrow \quad \quad \uparrow$
 $v_1 \quad \quad v_2$

$$\underline{A = U S V^T}$$

Exercises

Conceptual.

If A is an orthogonal matrix, what are the singular values of A ?

$$A^T A = I \quad !$$

$$\Rightarrow \sigma_i^2 = 1 \quad \text{for all } i$$

we choose the positive ones $\sigma_i = 1 \quad \forall_i$ (convention)

Exercises

Conceptual.

If $A = USV^T$, what is the SVD of A^T ?

$$A^T = (USV^T)^T = V S^T U^T = V S U^T$$

Furthermore, if A is invertible, what is the SVD of A^{-1} ?

$$A^{-1} = (USV^T)^{-1} = V S^{-1} U^T$$

$$S^{-1} = \begin{bmatrix} \frac{1}{\sigma_1} & & \\ & \ddots & \\ & & \frac{1}{\sigma_n} \end{bmatrix}$$